A Verbatim-Grounded, Field-Neutral Tag Layer for Cross-Disciplinary Reading of Academic Literature: A Proof of Format

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In three sentences. Researchers cannot find the work in other fields that studies the same thing as their own, because each field calls it by a different name. This paper describes a tag layer that gives every paper the same small set of tags and forces each tag to quote the exact sentence it came from — then reports honestly what that does and does not buy: free-text names for a phenomenon almost never collide across fields, while a controlled vocabulary of statistical terms does create real cross-field joins. The same instrument, read in reverse, can point at regions of concept space that the literature occupies but has never named.

A note on two numbers. The audited proof-of-format below is the frozen 126-paper corpus; the working corpus has since grown to 187 papers across 40 disciplines, including a digital-humanities batch added as a deliberate stress test (§8.1). Audited figures are not restated for the larger set, because they would have to be re-audited to be true.

Every paper gets the same small set of field-neutral tags, each forced to quote the sentence that licenses it — and the audit reports honestly what that buys: free-text names almost never join across fields, an audited statistical vocabulary does, and the same instrument, read in reverse, locates concepts the literature never named.

Summary view. Every paragraph of this paper has been replaced by one sentence stating what it argues; the figures and their legends are complete. The paragraphs themselves are in the full PDF deposited beside this file, and on the live page (shir-openu.github.io/meta-tagging-showcase/preprint.html), where clicking any summary opens the paragraph behind it.

Researchers cannot find the work in other fields that studies the same thing as their own, because each field calls it by a different name.

1. Motivation

Cross-field work fails not for lack of relevant results elsewhere but because every field names the same thing differently, and the existing structured resources index topics, which are field-specific by construction and therefore useless as join keys.

Describe every paper with the same small set of tags, and require each tag to quote the sentence it came from.

2. The idea

Each paper carries the same five field-neutral facets — claim type, phenomenon, claim relations, replication status, limitations — under one non-negotiable rule: every tag must carry the verbatim sentence that licenses it.

One graph: papers and tags as nodes, each tag-to-paper edge carrying the exact words that justify it.

2.1 The data structure: one property graph over a grounded tag layer

The layer is a three-level property graph: the raw corpus at the base, the grounded tag layer over it, and typed edges joining papers, tags, disciplines and canonical phenomena.

Two properties let it scale — verification becomes a set query instead of manual reading, and every facet value maps to a sorted postings list — but selective queries only avoid scanning unrelated records; they do not escape corpus size.

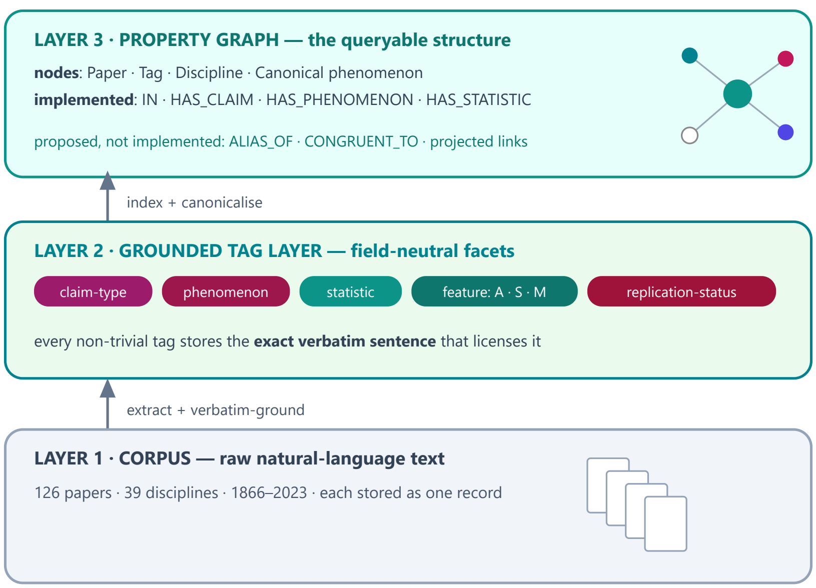

Fig. 1. The data structure: one **property graph** in three layers. A corpus of raw text (Layer 1) carries a grounded tag layer (Layer 2), which is read as a graph of nodes and edges (Layer 3) over which verification becomes a query. Full specification in the project's `GRAPH_SCHEMA`.

One JSON record per paper is the only source, and the build derives both classic structures — an inverted index and a typed adjacency list — from those records alone.

① one paper = one record (document store — a dict)

```
{
  "id": "newman2003-complex-networks",
  "discipline": "network-science", "year": 2003, "license": "arxiv-open",
  "title": "The Structure and Function of Complex Networks",
  "phenomenon": "complex networks (shared structure of real networks)",
  "content_tags": {
    "statistics": [{ "term":"correlation", "symbol":"r",
        "evidence":"For the data of Table III we find r = 0.621." }],
    "key_terms":  ["networks","degree","clustering","vertices","edges"] } }
```

② inverted index (facet → sorted postings) — expected O(1) key lookup

```
"statistic:correlation"      → [ 7, 12, 18, ... ] (39 papers · 22 disciplines)
"claim:method"               → [ 1, 3, 6, ... ]                      (55 papers)
"discipline:network-science" → [ 83, 84 ]                            (2 papers)
ALL = intersect; ANY = union; complement = universe difference
merge cost = Θ(Σ|P_i|) + output; worst case Θ(N)
```

③ implemented adjacency export (node → typed neighbours)

```
paper:newman2003 → [ IN:discipline:network-science,
                         HAS_STATISTIC:statistic:correlation, ... ]
not exported: ALIAS_OF · CONGRUENT_TO · projected links
```

Fig. 2. Implemented data structures. Each paper is one canonical JSON record. The build derives sorted postings and lazy metadata shards for search, plus a typed adjacency export. Dictionary lookup is expected O(1), but merging postings depends on their lengths and can be $\Theta(N)$; no corpus-size-independent complexity is claimed.

A curated corpus spanning dozens of disciplines, tagged to one schema, with every tag checked against the paper's own text.

3. Corpus and method

The corpus is a deliberately broad curated sample rather than a representative one, and the re-audit showed the intended character-level grounding was not consistently preserved, so grounding status is now an explicit field instead of an assumption.

Ten of 126 papers could not be faceted, every one because of a garbled text layer rather than resistant content — so that failure rate measures digitisation quality, not the tagging method.

Every open-access paper is rendered as a colour-tagged reading view so a human can audit each tag against the sentence that licenses it; paywalled papers are linked by DOI, never reproduced.

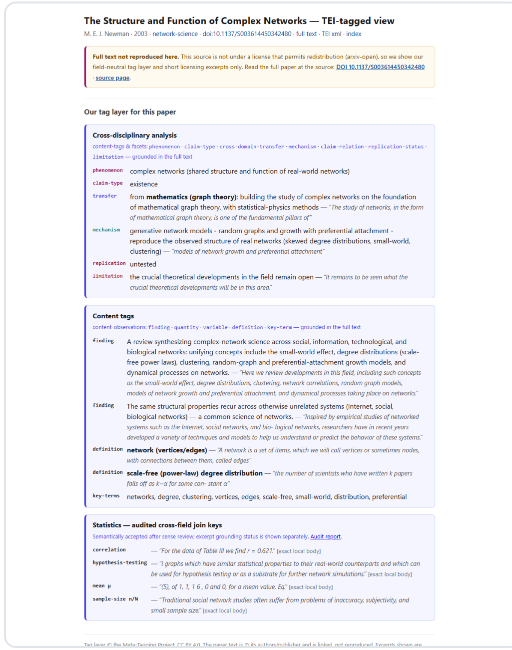

Fig. 3. The colour-tagged reading view (Newman 2003, an open-access paper): a colour legend plus the paper's own text with inline, type-labelled tags (persName, date, title, key-term, statistic…), each auditable against its sentence. **Click the figure** for the live page: shir-openu.github.io/meta-tagging-showcase/papers/newman2003….

The honest null result: free-text phenomenon names almost never collide across fields, so as join keys they fail.

4. Results

116 papers across 38 disciplines were fully faceted.

The central null result: written as free text the phenomenon strings almost never collide — exactly one pair of disciplines shares one verbatim — so what is missing is a canonical vocabulary, not more papers.

What does work as a cross-field join key is the statistics — the same terms recur in every discipline, so they connect papers that share nothing else.

5. Audited statistical terms as cross-field join keys

To test that fix directly we froze a 24-term statistical vocabulary, which travels across fields better than topic words but still marks a place to look, not proof that two papers did the same thing.

All 519 legacy paper–term assignments were re-audited against explicit written rules: 420 accepted, 99 rejected (19.1%), none left unresolved.

Because the legacy corpus preserved uneven provenance, the semantically accepted set and the stricter exact-local-body subset are reported side by side rather than quietly merged.

Across the full accepted set, at least one statistical key occurs in **100 of 126 papers** and **37 of 39 disciplines**. The strict exact-local-body subset contains **272 assignments in 77 papers across 32 disciplines**. The recalculated term-level results are:

Under audit the headline p-value and standard-deviation neighbourhood shrinks from 14 papers in 13 disciplines to 10 in 10, and to 3 in 3 under strict grounding.

The revised result is narrower than first claimed but the structural point survives: a controlled namespace produces more cross-field joins than highly specific free text.

A first structural test: the priming schema groups 16 papers across 8 disciplines under one canonical phenomenon even though each field names it differently.

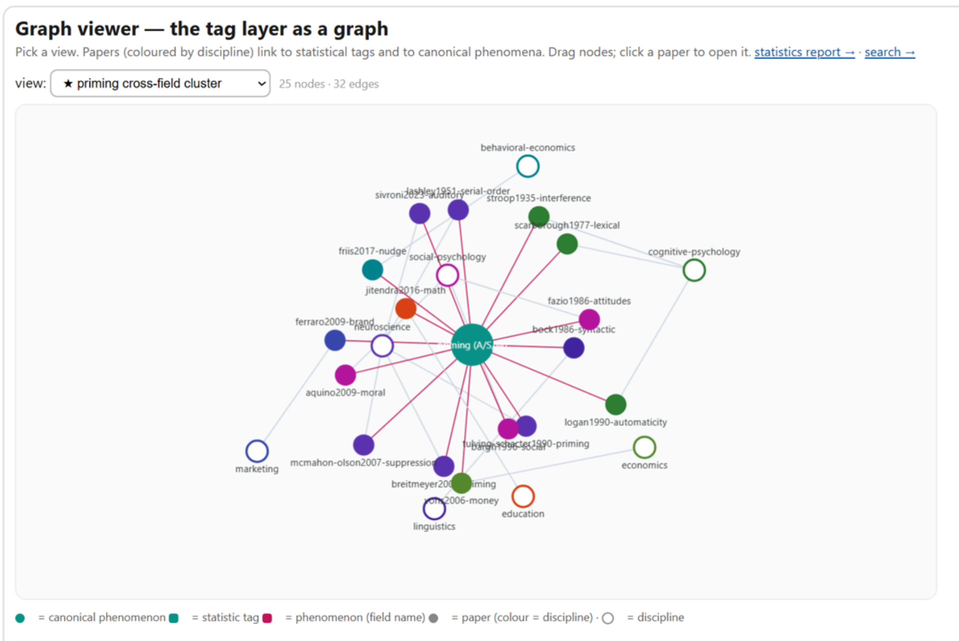

Fig. 4. The priming cross-field cluster in the interactive graph viewer (**real data**): one canonical *priming (A·S·M)* node at the centre links to 16 papers, coloured by discipline, wired out to their eight fields (neuroscience, cognitive and social psychology, economics, behavioral economics, marketing, education, linguistics) — though each field names it differently (interference, automaticity, nudge, money-priming, syntactic priming). 25 nodes · 32 edges. **Click the figure** for the live interactive graph: shir-openu.github.io/meta-tagging-showcase/GRAPH.html (opens locally as `GRAPH.html`).

We are careful about the claim: a shared *method* vocabulary is a weaker signal than a shared substantive finding — it marks where a connection *may* lie, consistent with the compass framing (§6), not a discovery.

Applied to priming across nine disciplines: association and modulation always hold, and only secondariness can fail.

5.1 A grounded test: does each field's priming actually carry the three features?

The non-trivial question is not whether priming recurs across fields — that is long known — but whether each field's priming actually carries all three defining features, grounded verbatim in that paper's own text.

Association and modulation hold in 16 of 16 papers; secondariness, the feature that can fail, is grounded in 14, and the two exceptions are genuine boundary findings.

- **Education** (schema-based mathematics instruction): the "prime" is a *deliberately taught* schema — the object of instruction, not an incidental cue — so secondariness is weak by construction.
- **Neuroscience** (serial-order motor control): motor *pre-activation* is a constitutive part of the produced action, not a task-irrelevant prime.

This is what grounding buys: the layer does not rubber-stamp the definition, it locates the boundary where a field stretches it, and the whole check is one tag intersection whose complement lists the exceptions automatically.

Fig. 5. Grounded A·S·M verification of the 16-paper priming cluster. Association and modulation are read from the structural slots (16/16); secondariness is a textual test that can fail (14/16). **Live interactive version** (per-paper evidence, drill-down): shir-openu.github.io/meta-tagging-showcase/PRIMING_SCALE.html.

At scale the same query draws one canonical priming node wired to its three features, each surrounding discipline coloured by whether all three hold or one is under-met.

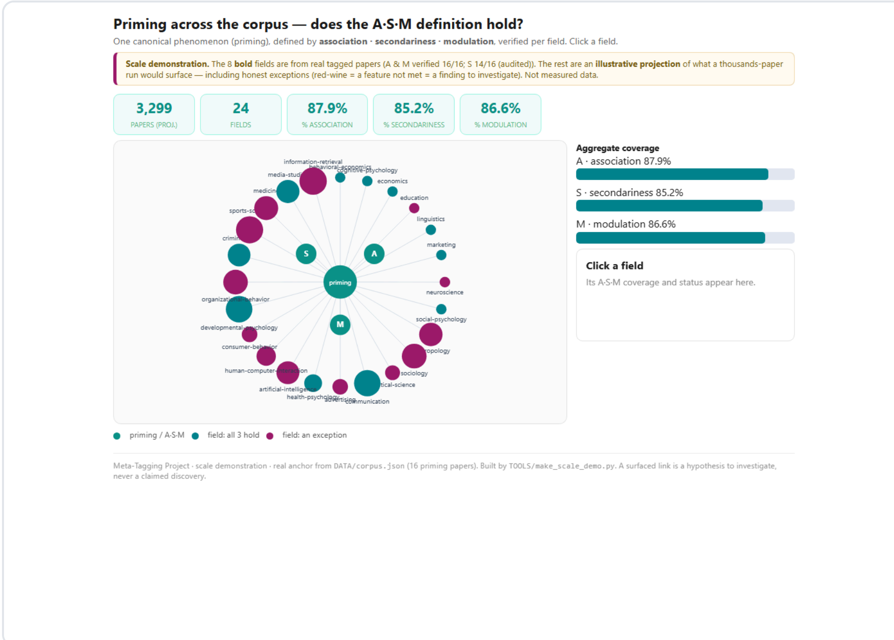

Fig. 6. The same A·S·M verification as an **illustrative projection** to a thousands-paper corpus (24 fields): one canonical *priming* node wired to its three features, each field turquoise if all three hold or red-wine if one is under-met (an exception to investigate). The eight bold fields carry the real audit numbers; the rest illustrate what a large run would surface — *not measured data*. **Click the figure** for the live interactive version: shir-openu.github.io/meta-tagging-showcase/PRIMING_SCALE.html.

A search over the tag layer that runs entirely in the browser, with no server, so the demonstration cannot rot.

5.2 Querying the layer: an implemented static, client-side search

The search is plain static files — a manifest, a postings dictionary and lazy shards — so anyone can query the facets across all fields at once with no server and no database behind it.

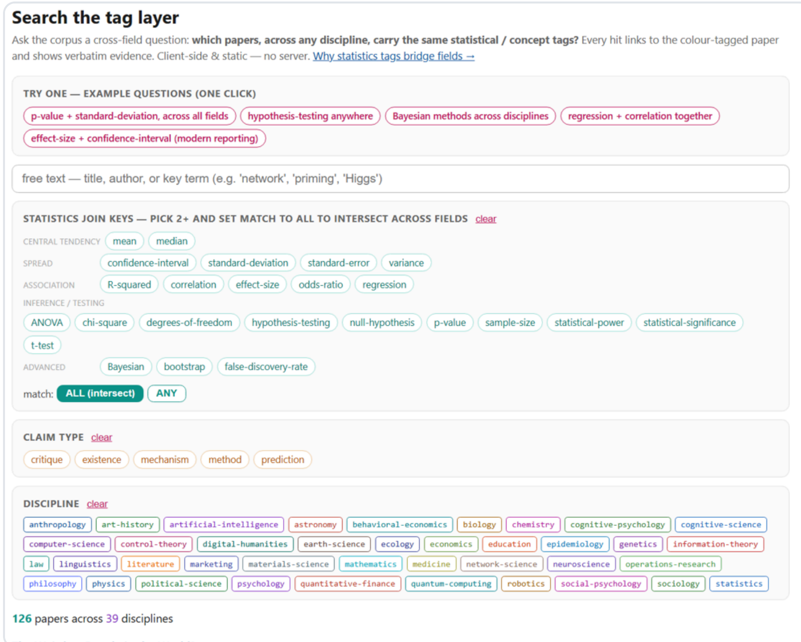

Fig. 7. The static, client-side search. The reader intersects field-neutral tags — statistics join-keys (grouped), claim-type, discipline, free text — with an `ALL` / `ANY` switch; results are papers from any field, each with its matched tags, key terms, and a link to the colour-tagged evidence. No application server; static HTTP hosting is required. **Click the figure** for the live search: shir-openu.github.io/meta-tagging-showcase/SEARCH.html.

The same instrument that fails to find shared names can locate regions the literature occupies but never named — demonstrated for short-term implicit memory.

5.3 The negative space: locating terms the literature lacks

Read the other way round, the null result becomes a detector: a region of concept space that is populated yet shares no label is evidence the literature never coined a term for it — something no citation graph can show, because it operates on links between papers, not on the words they failed to share.

The procedure is to cross two established dimensions and look for cells that are populated but unnamed, claiming absence only where the region is demonstrably occupied.

The memory taxonomies cross duration against awareness asymmetrically, leaving the cell that would be called short-term implicit memory crowded with data and almost without a name.

About 4,350 works occupy that region under five paradigm-specific names against roughly 52 under any unifying one: the concept is not absent but fragmented, which is why a researcher working in one paradigm has no name to search for the others by.

This establishes that the gap is locatable and countable rather than impressionistic — not that the missing term ought to be adopted, and one worked case is a method with an example, not a survey.

Two separate claims: the worked case above rests on its own evidence, while the corpus-wide sweep that follows is a weaker exercise which can come back small without touching it.

Checked against Wikidata on three independent probes, only three of the corpus's own multi-word key phrases are genuinely absent — lexical decision, category accessibility and repetition suppression — so the earlier count of fourteen is corrected down, and the reduced figure is the one claimed.

Sharper than absence: five phrases survive only as redirects — social, semantic and conceptual priming all collapse into a single Wikidata item — which is the argument for linking out to authorities rather than being replaced by them.

This is a compass, not an answer: it points at where to look, and does not claim to settle what is found there.

6. Framing: a compass, not an answer

The layer is a compass, not an answer engine: it points at where a cross-field connection may lie so a scholar knows where to spend the slow close reading that alone can establish it.

Because reading on demand does not scale, and an ungrounded summary cannot be checked by the person who did not do the reading.

6.1 Why a grounded layer, rather than reading on demand

If a model can already read any of these papers on demand, what is a tag layer for? Not access — these papers were always available — but four properties that reading on demand does not have.

- **Verifiability, not access.** A model's claim about a paper cannot be checked without redoing the reading, whereas every assignment here carries the sentence that licenses it — so a reader who doubts a judgement can inspect the quote and disagree on the record.
- **Questions about the set, not the paper.** “Which papers, in any field, carry both of these terms?” is a set operation over an index, not a question you can put to a reader one paper at a time.
- **Reproducibility.** A layer is computed once and queried repeatedly with the same answer every time, so a disagreement about a number is settled by re-running the build rather than by trusting the author.
- **Join keys must be constructed, not assumed.** A model trained on the literature inherits the literature's vocabulary and is therefore blind to the words it never coined — you cannot ask an on-demand reader to find a term that does not exist.

The fifth reason is human: the obstacle is rarely that a literature is inaccessible but that it is unrecognisable, and a shared notation makes the common structure visible — offered as design rationale, not as a measured claim about readers.

Stated plainly: a curated corpus, single-coder tagging in places, a measured κ of 0.767 on the statistical layer only, and facets that are not all grounded.

7. Limitations

- The statistic facet records an accepted statistical occurrence, not necessarily a method the paper used, so a shared key marks a candidate neighbourhood rather than a shared result.
- Evidence provenance is incomplete: only 272 of the 420 accepted statistical assignments have an exact excerpt located in a locally stored article body, and the rest must never be described as exact-body verified.
- Tagging currently depends on a large language model to read prose; the grounding guard bounds but does not eliminate interpretation error.
- The re-audit's reliability is measured rather than asserted: a blinded second coder re-coded 120 assignments for $\kappa = 0.767$, and the disagreement was asymmetric in the direction of this audit being the stricter of the two.
- The corpus is curated for breadth, not sampled representatively; distributions above describe *this* corpus only.
- Not every facet is verbatim-grounded — the concepts facet is an analyst's label with no quote behind it — which is why §5.3 was re-run on key phrases and its claim reduced.
- The place field turned out to hold the authors' institutional city rather than the publisher's, and two of its values are not places at all; the reconciliation records that instead of quietly cleaning it.
- The study-location facet is grounded but thin — 4 of the 38 human-subjects papers state it in a sentence a machine can find — and the ceiling is structural, because most of the relevant literature here is paywalled and never published.
- Replication status and limitations are taken only from each paper's own words; known external replication failures are deliberately not injected, because nothing in the source grounds them.
- The phenomenon join key needs canonicalisation before it can surface cross-field links at scale (§4).
- The implemented graph export does not include `ALIAS_OF`, `CONGRUENT_TO`, projected links, or graph-analysis results. The priming projection is a separate curated analysis and Fig. 6 is explicitly illustrative.
- Full-text HTML in Git is the wrong store beyond this scale: production should version metadata and code, put redistributable full text in object storage, and serve postings and record shards statically.
- Novelty claims are unfalsifiable in the strict sense; we can search and report, not prove a negative.

TEI, Web Annotation, nanopublications and distant reading each do part of this; what is new is requiring all of it at once.

8. Relation to prior work

It sits among TEI, Web Annotation, nanopublications, knowledge graphs and distant reading; what is new is the union of mandatory verbatim grounding, field-neutral facets, and a claim about cross-field links that was actually tested.

A gazetteer reconciles a place name across languages; this layer reconciles a sentence across disciplines — the same operation, so the same standards fit.

8.1 The same operation across languages and across disciplines

The closest working analogue is not text mining but a historical gazetteer: discipline is to a tag layer what language is to a gazetteer — a surface form, an authority record, and a pointer back to where the form was found.

Two existing standards already do this work — SKOS for the vocabularies, and the Web Annotation TextQuoteSelector, which turns out to be this project's house rule on verbatim grounding, already standardised by the W3C.

Reconciliation to Wikidata succeeded completely for the structural facets (39/39 disciplines, 13/13 methods) and only partly for the substantive ones (15/25 statistics, 61/88 concepts) — an asymmetry that runs along the very gradient this layer was built for.

Converting the tags to annotations re-verified them independently of our own checker: 99.4% of publishable spans anchor exactly, once ellipsis-joined quotes, ligature and whitespace drift, and our own tag chips cutting quotes in half were fixed.

The analogy stops at entities: a place match can be scored against ground truth, a concept match cannot — which is exactly why this layer demands a quote for every tag and reports κ instead of precision, and why 42 ambiguous place names had to be decided by hand.

Nine digital-humanities and Linked-Open-Data papers stress-tested the facets: they mostly held, with one clean failure — the method vocabulary fitted none of 4 of the 9 — reported as a gap rather than papered over by stretching a label.

One seminar paper contradicts a decision made here: Münz-Manor and Marienberg-Milikowsky began with no tagset at all and let visualisation reshape their categories, so their tagset is an output of the analysis rather than an input to it.

This layer freezes its facets in advance, and the honest reading is that the two choices answer different questions: a scheme that emerges fits its own corpus best, and only a fixed one can serve as a join key across corpora never compared before.

Dekel and Marienberg-Milikowsky attack distant reading at the same joint: they replace the authoritative summary with many people's readings, we replace it with what each paper itself says, quoted — two refusals of the same shortcut.

MEHDIE reports the same shape of gap on a different axis — even Wikidata has low recall for the Semitic-script toponyms their sources use — which argues for domain-grounded layers that link out to central authorities instead of waiting for them.

Everything is public and rebuildable: corpus, code, vocabulary, annotations, and a rights manifest saying what may be redistributed and what may not.

9. Availability

The public showcase carries the report and, wherever redistribution rights were verified per paper, the full colour-tagged text; the repository adds the canonical corpus, the audit decisions, invariant validation and a one-command reproducible build.

The layer is also published in two standards it did not invent — SKOS and W3C Web Annotation, about 47,800 triples — with the reliability layer deliberately never exported and every asserted link carrying its own evidence.

This work grew out of the Beyond Words DHSS summer school, whose organisers, instructors and participants seeded the idea of a single uniform tag layer; it was carried out independently, without funding.

1. F. Moretti, *Distant Reading*, 2013 — the "distant reading" framing this work adapts as "far reading."
2. W3C, *Web Annotation Data Model*, Recommendation, 2017 — stand-off annotation with explicit targets.
3. P. Groth, A. Gibson, J. Velterop, "The anatomy of a nanopublication," 2010 — minimal machine-readable assertions.
4. S. Rusinek, T. Sagi, M. Zaga, E. Lev, M. Lavee, "MEHDIE: The Middle East Heritage Data Integration Endeavor," *Digital Humanities 2023*, Graz (CC BY 4.0); extended as T. Sagi et al., *Digital Scholarship in the Humanities* 41(Suppl. 1), i238–i246, 2026, doi:10.1093/llc/fqaf111 — reconciliation across languages rather than disciplines.
5. S. Isaj, E. Zimányi, T. B. Pedersen, "Multi-Source Spatial Entity Linkage," *SSDBM*, 2019, doi:10.1145/3340964.3340979 — entity linkage scored against labelled data, the contrast case for §8.1.
6. W3C, *SKOS Simple Knowledge Organization System Reference*, Recommendation, 2009 — the concept-scheme model used for the vocabulary export.
7. O. Münz-Manor, I. Marienberg-Milikowsky, "Visualization of Categorization: How to See the Wood and the Trees," *Digital Humanities Quarterly* 17(3), 2023 — a tagset that emerged from the analysis rather than preceding it.
8. Y. Dekel, I. Marienberg-Milikowsky, "From Distant to Public Reading. The (Hebrew) Novel in the Eyes of Many," *magazén* 2(2), 225–252, 2021, doi:10.30687/mag/2724-3923/2021/01/003 — distant reading without the authoritative history.

Preprint · v11 linked-open-data revision · July 2026 · licensed CC BY 4.0 · DOI: 10.5281/zenodo.21432188 · cite via CITATION.cff in the project repository. The v8 audit re-audited all 519 legacy statistical assignments, recalculated §5, separated semantic acceptance from exact-local-body grounding, implemented a sharded sorted-postings search and reproducible build pipeline, exported only supported graph relations, and corrected the architecture and complexity claims. v9 adds a per-paper redistribution-rights manifest, states the audit method (rule-based with LLM-assisted adjudication) and the absence of an inter-rater study, and reconciles all release metadata. Earlier versions introduced the priming A·S·M analysis and the statistical join-key experiment. v11 reconciles the controlled vocabularies to Wikidata, publishes the layer as SKOS and W3C Web Annotation (§8.1), re-verifies 99.4% of verbatim spans by anchoring them as TextQuoteSelectors, adds nine digital-humanities papers as a facet stress test, adds a grounded study-location facet, and reduces the §5.3 negative-space claim to what the papers' own words support. v17 tags five papers from the digital-humanities summer school this

work comes out of, and records a methodological disagreement with one of them rather than smoothing it over: their tagset emerged from the analysis, ours is frozen in advance, and §8.1 states what each choice buys and costs. v19 replaces every paragraph with one sentence written to say what that paragraph argues, and puts the paragraph itself one click behind it; the figures and their legends are always shown, tables open on request, and the summaries are kept in a separate file so that a paragraph whose text changes loses its summary rather than keeping a stale one.